DELIVERY ROUTE OPTIMIZER

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The **Delivery Route Optimizer** is a smart logistics solution designed to improve the efficiency of daily delivery operations. The system processes raw and inconsistent address data by cleaning, validating, and standardizing delivery locations to ensure accurate route planning. It intelligently assigns delivery stops to available vehicles based on factors such as vehicle capacity, delivery demand, and operational constraints. Using route optimization algorithms, the system generates near-optimal delivery routes that minimize total travel distance, delivery time, and fuel consumption while considering time windows and real-world limitations. Since finding the exact optimal solution becomes computationally expensive for large-scale delivery networks, the proposed solution employs efficient approximation techniques to produce high-quality results within practical time limits. The system also provides detailed cost analysis, route quality metrics, and performance reports, enabling logistics operators to evaluate efficiency and make informed decisions. Overall, the Delivery Route Optimizer enhances productivity, reduces operational costs, and improves the reliability of delivery services.

Signature of the Guide